GOOG EWMA Volatility and American Call-Option Pricing

Nasdaq historical data, EWMA volatility estimation and a 100-step binomial tree

Prepared by Kan Wai Kit | Valuation date: 20 October 2026

Executive summary

This project downloads GOOG historical prices from Nasdaq, estimates historical and exponentially weighted volatility, and prices an American call option with a Cox-Ross-Rubinstein-style binomial tree. Using a spot price of $336.40, strike price of $225.00, 84 calendar days to maturity and EWMA annualized volatility of 22.1440%, the model produces an option value of $113.1705. The contract is deep in the money, so most of its value is intrinsic value rather than time value.

1. Objective and data

The objective is to combine market-data processing, volatility estimation and derivative valuation in one reproducible Python workflow. Historical GOOG data are requested from the Nasdaq historical-price endpoint and cleaned before return and volatility calculations.

- Underlying: GOOG common stock.
- Historical sample: 21 October 2021 to 20 October 2026.
- Number of cleaned daily observations: 1,254.
- Latest closing price used as spot: $336.40.
- Output workbook: EWMA_Volatility.xlsx.

2. Data preparation

The API response is converted from JSON into a pandas DataFrame. Date, close, open, high, low and volume fields are renamed and converted into appropriate numeric or datetime formats. Invalid dates, missing closing prices and non-positive prices are removed, after which the observations are sorted chronologically.

Data item	Result
First observation	21 October 2021
Latest observation	20 October 2026
Cleaned observations	1,254
Latest GOOG close	$336.40

3. Volatility estimation

Historical volatility

Daily continuously compounded returns are calculated as the natural logarithm of the ratio of consecutive closing prices. The sample standard deviation of daily log returns is annualized using the square root of 252 trading days.

Historical annualized volatility = 31.9619%.

EWMA volatility

The exponentially weighted moving average model assigns greater weight to recent squared returns. A decay factor of 0.94 is used, so the newest squared return receives a weight of 0.06. The latest daily EWMA variance is converted to daily volatility and then annualized by multiplying by the square root of 252.

Volatility measure	Estimate	Interpretation
Historical annualized volatility	31.9619%	Average variability across the full five-year sample.
Latest EWMA annualized volatility	22.1440%	Current volatility estimate with greater emphasis on recent returns.
Difference	-9.8179 percentage points	Recent volatility is below the full-sample historical estimate.

The lower EWMA estimate indicates that recent GOOG price movements were less volatile than the broader five-year history. The EWMA estimate is therefore used as the option-pricing volatility input because it responds more quickly to current market conditions.

4. Binomial option-pricing model

The project uses a 100-step recombining binomial tree. At each step, the stock price can move up or down by factors determined by the EWMA volatility. Risk-neutral probabilities discount expected option payoffs using continuously compounded rates. Because the contract is specified as American, continuation value is compared with immediate exercise value at every node.

Input	Value
Option type	American call
Spot price, S	$336.40
Strike price, K	$225.00
Continuously compounded risk-free rate, r	3.8259%
Continuously compounded dividend yield, q	0.2613%
Time to maturity, T	0.230137 years / 84 days
Annualized volatility, sigma	22.1440%
Tree steps	100

5. Valuation result and interpretation

Measure	Result
Model option price	$113.1705

Immediate exercise value	$111.4000
Estimated time value	$1.7705
Spot-to-strike ratio	1.4951x
Option premium as % of spot	33.64%

The call is deep in the money because the spot price exceeds the strike by $111.40. The binomial price exceeds intrinsic value by approximately $1.7705, representing the estimated time value of the remaining 84-day option life. This relationship is economically sensible because an option should not be worth less than its immediate exercise value in the American framework.

The model price is driven primarily by intrinsic value. Changes in volatility, time to maturity, interest rates and dividends mainly affect the smaller time-value component for this deep-in-the-money contract.

6. Model controls and limitations

- The code checks that the risk-neutral probability lies between zero and one; otherwise model inputs or the number of steps must be reviewed.
- The Nasdaq close field should be checked for corporate-action adjustment. Unadjusted stock splits or special distributions could distort returns and volatility.
- The dividend assumption is represented as a continuous annual yield derived from four payments of $0.22. Actual dividend timing is discrete, so a discrete-dividend tree would provide more precise American exercise analysis.
- The risk-free rate, dividend assumption, strike and 84-day maturity are user inputs rather than automatically matched to a specific listed option contract.
- The EWMA decay factor of 0.94 is a modelling choice. Sensitivity analysis should compare alternative decay factors and volatility measures.
- A 100-step tree reduces discretization error but does not eliminate it. Convergence should be tested using larger step counts such as 200, 500 and 1,000.

7. Conclusion

The project demonstrates an end-to-end workflow from a live market-data endpoint to volatility estimation and option valuation. Historical annualized volatility is 31.9619%, while the latest EWMA estimate is lower at 22.1440%, indicating calmer recent price behavior relative to the full sample. With the stated option inputs, the 100-step American binomial tree values the call at $113.1705. The result is consistent with a deep-in-the-money option: $111.40 of value is intrinsic and approximately $1.7705 is time value.